

What Are You Proving?

1. Proof Statement:

My portfolio demonstrates one core capability: I build practical AI applications that address real-world problems by applying Machine Learning, Generative AI, Large Language Models (LLMs), and modern AI engineering practices. It is designed for AI recruiters and hiring managers looking for an AI Engineer or Generative AI Developer who can research real-world challenges, design practical AI systems, and deliver end-to-end solutions through internships and technical projects. My goal is for recruiters to confidently contact me for an AI internship or entry-level AI engineering opportunity after reviewing my portfolio.

2. Why This Needs to Exist:

My resume and LinkedIn highlight my education, skills, certifications, and internships, but they cannot fully demonstrate how I approach problems, make engineering decisions, and build practical AI applications. My portfolio provides evidence of my technical thinking, implementation process, and problem-solving approach through real projects and internship work.

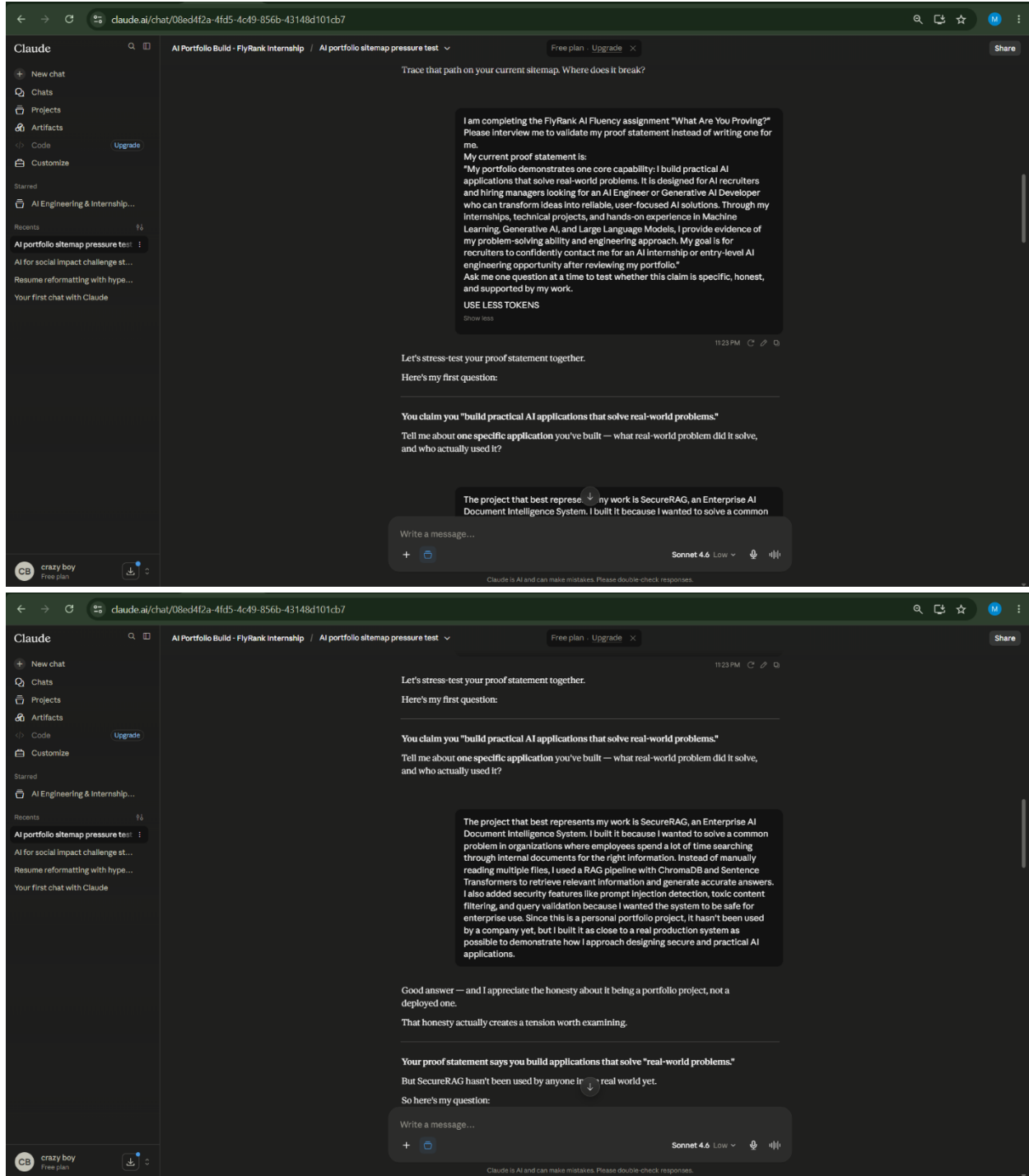
3. AI Reflection

To refine my proof statement, I used Claude as a thinking partner rather than an author. Through an interview-style conversation, Claude challenged my assumptions, asked me to justify my claim with evidence from my projects and internships, and helped me narrow my statement to accurately reflect what I have genuinely demonstrated. This process improved the clarity, honesty, and credibility of my proof statement.

4. Claude Feedback Summary

During the interview, Claude challenged my original proof statement by asking me to justify my claims with evidence from my projects and internships. Through this discussion, I realized that saying I "solve real-world problems" overstated the current stage of some of my projects, as they are research-informed prototypes rather than production deployments. Based on the feedback, I refined my proof statement to say that I build practical AI applications that address real-world problems. Claude also confirmed that my projects and internships demonstrate my ability to research domain problems, design practical AI systems, understand technical trade-offs, think from the user's perspective, and deliver end-to-end AI solutions in both independent and team environments. This process helped me create a proof statement that is specific, honest, and fully supported by my experience.

5. Screenshot of Claude Interview



claude.ai/chat/08ed4f2a-4fd5-4c49-856b-43148d101cb7

claude

New chat

Chats

Projects

Artifacts

Code

Customize

Upgrade

Starred

AI Engineering & Internship...

Recents

AI portfolio sitemap pressure test

AI for social impact challenge st...

Resume reformatting with hype...

Your first chat with Claude

AI Portfolio Build - FlyRank Internship

AI portfolio sitemap pressure test

Free plan

Upgrade

Share

by a company yet, but I built it as close to a real production system as possible to demonstrate how I approach designing secure and practical AI applications.

Good answer — and I appreciate the honesty about it being a portfolio project, not a deployed one.

That honesty actually creates a tension worth examining.

Your proof statement says you build applications that solve "real-world problems."

But SecureRAG hasn't been used by anyone in the real world yet.

So here's my question:

What specific evidence do you have that the problem you identified — employees wasting time searching internal documents — is a real, validated problem?

Did you talk to anyone who experiences it? Read industry research? Or did you assume it was a problem worth solving?

I didn't identify this problem by interviewing enterprise users directly, so I don't want to claim that. I identified it by studying how Retrieval-Augmented Generation (RAG) is used in enterprise AI systems and by reading documentation, technical blogs, and case studies from companies building AI assistants for internal knowledge management. A common challenge mentioned across these resources is that employees spend significant time searching through scattered documents, which affects productivity. I built SecureRAG to address that well-documented problem while also learning how to design a secure, enterprise-style RAG application. So the problem is real, but my project is a technical demonstration of one possible solution rather than a validated product.

That's one of the most honest and well-reasoned answers I've heard from a student. You just

Write a message...

Sonnet 4.6 Low

Claude is AI and can make mistakes. Please double-check responses.

claude.ai/chat/08ed4f2a-4fd5-4c49-856b-43148d101cb7

claude

New chat

Chats

Projects

Artifacts

Code

Customize

Upgrade

Starred

AI Engineering & Internship...

Recents

AI portfolio sitemap pressure test

AI for social impact challenge st...

Resume reformatting with hype...

Your first chat with Claude

AI Portfolio Build - FlyRank Internship

AI portfolio sitemap pressure test

Free plan

Upgrade

Share

real, but my project is a technical demonstration of one possible solution rather than a validated product.

That's one of the most honest and well-reasoned answers I've heard from a student. You just drew a clear line between "real problem" and "validated solution" — that's mature engineering thinking.

Now let's push on the technical side.

You mentioned security as a key differentiator — prompt injection detection, toxic content filtering, query validation.

How did you actually implement prompt injection detection? Walk me through the technical approach, not just what it does.

I implemented prompt injection detection as the first security check before the user's query entered the RAG pipeline. Whenever a user submits a query, I first normalize the input by converting it to lowercase and cleaning unnecessary whitespace. Then I compare it against a predefined list of prompt injection patterns and suspicious phrases such as "ignore previous instructions," "reveal the system prompt," or "bypass security." If any of these patterns are detected, the request is blocked before it reaches the retrieval or generation stages. At the same time, the system records the event in an audit log with the timestamp, query, and security status. I chose this rule-based approach because it is simple, transparent, and easy to maintain. Although it cannot detect every advanced attack, it provides a strong first layer of defense. In the future, I would like to enhance it using an LLM-based or machine learning-based detection model to identify more sophisticated prompt injection attempts.

Show less

11:29 PM

Solid explanation — you know what you built — why you made those tradeoffs. Naming the limitation unprompted shows good engineering judgment.

Write a message...

Sonnet 4.6 Low

Claude is AI and can make mistakes. Please double-check responses.

← → ↺ 🌐 claude.ai/chat/08ed4f2a-4fd5-4c49-856b-43148d101cb7 🔍 📄 ⭐ 🌐

Claude 🔍 📄

➕ New chat

🗉 Chats

📁 Projects

📎 Artifacts

🔗 Code Upgrade

⚙️ Customize

Started

📁 AI Engineering & Internship...

Recents 📄

AI portfolio sitemap pressure test: 📄

AI for social impact challenge st...

Resume reformatting with hype...

Your first chat with Claude

AI Portfolio Build - FlyRank Internship / AI portfolio sitemap pressure test 🔽 Free plan · Upgrade ✕ Share

11:29 PM 🔄 📄 🗉

Solid explanation — you know what you built and why you made those tradeoffs. Naming the limitation unprompted shows good engineering judgment.

Now one harder question:

You chose a rule-based pattern-matching approach over an ML-based one.

A recruiter or technical interviewer might ask: "Couldn't a bad actor simply rephrase the attack to avoid your pattern list?"

How would you defend that design decision?

I would agree with the interviewer that a rule-based approach can be bypassed if someone rephrases an attack. I never considered it a complete security solution. My decision was based on the scope of the project. I wanted to build a security layer that was simple, explainable, and easy to test before adding more complex detection methods. For a portfolio project, this allowed me to demonstrate the overall security workflow—detecting suspicious input, blocking unsafe requests, and logging security events—without introducing unnecessary complexity. If I were building this for a production environment, I would combine rule-based filtering with an LLM-based guardrail or a machine learning classifier to detect semantic prompt injection attacks that don't rely on fixed keywords. I see the rule-based layer as the first line of defense, not the only one.

11:30 PM 🔄 📄 🗉

That's a strong, honest answer. You defended a tradeoff without overselling it — exactly what good engineers do.

You've now demonstrated:

- A real problem backed by research

Write a message...

+ 🗉 Sonnet 4.6 Low 🔊 🗑️

Claude is AI and can make mistakes. Please double-check responses.

← → ↺ 🌐 claude.ai/chat/08ed4f2a-4fd5-4c49-856b-43148d101cb7 🔍 📄 ⭐ 🌐

Claude 🔍 📄

➕ New chat

🗉 Chats

📁 Projects

📎 Artifacts

🔗 Code Upgrade

⚙️ Customize

Started

📁 AI Engineering & Internship...

Recents 📄

AI portfolio sitemap pressure test: 📄

AI for social impact challenge st...

Resume reformatting with hype...

Your first chat with Claude

AI Portfolio Build - FlyRank Internship / AI portfolio sitemap pressure test 🔽 Free plan · Upgrade ✕ Share

as the first line of defense, not the only one.

That's a strong, honest answer. You defended a tradeoff without overselling it — exactly what good engineers do.

You've now demonstrated:

- A real problem backed by research
- A clear technical implementation
- Honest tradeoff reasoning

Now let's test the "reliable, user-focused" part of your proof statement.

You built SecureRAG to be enterprise-ready. But who was the "user" in your mind when designing it?

Did you make any specific design decisions because of that user — or did you build what made technical sense and label it enterprise-ready afterward?

When I designed SecureRAG, I had two primary users in mind: employees who need quick access to information from internal documents and administrators who are responsible for maintaining security and compliance. Those users influenced several design decisions. For employees, I focused on making the system return relevant answers quickly through semantic search so they wouldn't have to manually search multiple documents. For administrators, I implemented prompt injection detection, toxic content filtering, query validation, and audit logging so every interaction could be monitored and reviewed. I didn't conduct formal user testing, so I can't claim the design has been validated by real enterprise users. Instead, I designed the system by studying common enterprise AI workflows and security requirements. My goal was to build a realistic prototype that balances usability with security rather than simply adding enterprise features after the

Write a message...

+ 🗉 Sonnet 4.6 Low 🔊 🗑️

Claude is AI and can make mistakes. Please double-check responses.

Claude

New chat

Chats

Projects

Artifacts

Code

Customize

Starred

AI Engineering & Internship...

Recents

AI portfolio sitemap pressure test

AI for social impact challenge st...

Resume reformatting with hype...

Your first chat with Claude

AI Portfolio Build - FlyRank internship / AI portfolio sitemap pressure test

Free plan · Upgrade

Share

the design has been validated by real enterprise users. Instead, I designed the system by studying common enterprise AI workflows and security requirements. My goal was to build a realistic prototype that balances usability with security rather than simply adding enterprise features after the implementation.

Show less

11:31 PM

That's a well-structured answer. Defining two distinct user types and connecting each to specific design decisions shows systems thinking, not just coding.

We've now stress-tested your proof statement across four dimensions. Here's what the evidence supports and what it doesn't:

Claim	Verdict
"Build practical AI applications"	Supported
"Solve real-world problems"	Partially — problem is real, solution is unvalidated
"Reliable, user-focused solutions"	Supported with honest caveats
"Internships, technical projects, hands-on experience"	Untested — we only examined one project

Before I give you final feedback, one last question:

Your proof statement mentions internships as evidence. What did you build or contribute during an internship that supports the same claim — that you can turn ideas into working AI solutions?

One example is my AI and Machine Learning internship at Elevates Lab.

Write a message...

Sonnet 4.6 Low

crazy boy

Free plan

Claude is AI and can make mistakes. Please double-check responses.